

School of Electrical and Computer Engineering

Course Title: Fast Robots

Author: E. Farrell Helbling, ECE

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Credit Hours: 4 hours

Catalog Description:

The course focus is on systems level design and implementation of dynamic autonomous robots. With the recent DIY movement, design of kinematic robots is largely becoming a software challenge. In dynamic robots, however, any latency or noise can be detrimental. We will design a fast autonomous car, explore dynamic behaviors, acting forces, sensors, and reactive control on an embedded processor, as well as the benefit of partial off-board computation. Students will learn how to derive design specifications from abstract problem descriptions and gain familiarity with rapid prototyping techniques, system debugging, system evaluation, and online dissemination of work.

Course Frequency:

Every Spring.

Prerequisites:

ECE3400 or MAE3780, and ECE3100 or equivalent.

Beneficial: ECE3140, MAE4180/CS3758/ECE4180, MAE4810, and CS4750/ECE4770/MAE4760.

Corequisites:

None

Student Preparation Summary:

This class complements MAE4810 Robot Perception and CS3758/ECE4180/MAE4180 Autonomous Mobile Robots and is intended for students across CoE. Programming experience in C/C++ and Python is required. Knowledge of some or all of the following topics is beneficial: robotics, control, rapid prototyping, digital signal processing, and embedded systems.

Textbook(s) and/or Other Required Materials:

The course material is based on lectures and freely available material. All students will receive a hardware kit sponsored by ASML to complete labs. Students can use computers in PH427 to complete software assignments. The software is also available for installation on private laptops (note that this software may not work on most tablets).

ECE Open CourseWare Link: N/A

Class and Laboratory Schedule:

Lectures: Two 75min lectures per week; students are required to partake.

Recitations: None required.

Labs: Hardware implementation, debugging, and evaluation is involved. Students are required to partake, in person, in a 2hrs 30min lab session each week.

Final Showcase: All students must partake in a final robot showcase on May 5th, 2026. Students must notify the instructor early about scheduling constraints that only permit partial participation.

Note: Syllabus subject to change prior to course start.

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Assignments, Exams and Projects:

Homework: Homework is composed of labs and lab write-ups.

Exams: There are no dedicated exams

Design Projects: Weekly labs themed around dynamic robots including robust design, perception, reactive control, planning, and communication will guide students towards a real robot which can perform students, map out its environment, localize within it, and plan and execute a path.

Course Grading Scheme: 10% participation, 60% labs, 30% write-ups.

5000-level student reports are evaluated against others on the same level, i.e. separate from 4000-level student reports.

Note that the 5000-level version of this class involves additional (and more challenging) assignments in half the labs, including: Lab 1) programming an interactive routine to make the controller react upon a whistle; Lab 2) measuring the effective data rate and reasoning about reliability of Bluetooth data transfer; Lab 3) computing and demonstrating the ability to measure magnetic north from magnetometer outputs; Lab 4) considering the appropriate frequency of embedded PWM library functions to drive particular actuators and characterizing the actual range of PWM-values for which the car reacts; Lab 5) Discussing why and show how even small initial errors can lead to large problems in PID proximity control.

Details List of Topics Covered:

- Systems design process (translate demands and constraints, sketching, defining interfaces)
- Programming in C and Python, for efficient control
- Sensors (passive/active), signal processing, filtering
- Mechanical components (gears, axels, forces)
- Robot forward kinematics and dynamics
- Communication (Bluetooth, i2c, serial, protocols)
- Power (batteries)
- Planning and search-algorithms
- Tools* (CAD, Unity, soldering, wiring, laser cutting, 3D printing, hand tools, molding)

*Subject to change depending on access.

Student Outcomes [ABET]:

Students will learn how...

- ...to translate abstract engineering challenges to practical design specifications
- ...to reason about the practicality of components given real-world constraints
- ...to design systems consisting of electronics, software, and mechanics
- ...to robustly integrate systems operating in the real world
- ...how to disseminate their work to their peers and an engineering audience
- ...to predict the likely social and environmental effects of their design

Academic Integrity:

Students are expected to abide by the Cornell University Code of Academic Integrity with work submitted for credit representing the student's own work.

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